

Rural Conflict and Urban Markets: Evidence from Southern Chile

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Rural conflict can weaken nearby urban markets even when events occur outside the city. I study this spillover in southern Chile by linking geocoded Mapuche conflict events to tax records, satellite lights, fiscal appraisals, cadastre, construction, and fixed urban census-block panels. The design measures annual nearby conflict exposure for fixed urban markets and, separately, exposure differences across blocks within the same market. Greater external conflict exposure is associated with lower tax-reported sales and fewer registered enterprises. The association does not appear as a uniform contemporaneous decline in nighttime lights, fiscal appraisals, or new built area. Continuous lead-lag profiles place the enterprise association at longer lags and describe later movements in some appraisal measures, but pre-exposure movement in several series limits a causal timing interpretation. Complementary staggered-DiD profiles around conflict spikes place the clearest medium-run patterns in enterprise activity, with delayed declines in appraisal measures and nighttime lights in some specifications. The block-level design does not show a broad contemporaneous reallocation of lights, assessed values, new built area, or commercial use within urban markets. The evidence points to a bounded association between nearby conflict and municipal formal activity, rather than to a general contraction inside the fixed urban footprint. These results imply that unresolved territorial conflict can impose economic costs beyond the actors and places directly involved, making conflict resolution and institutional clarity relevant for nearby urban development as well as for the rural sites where disputes occur.

Keywords: Indigenous conflict; political violence; local economic activity; nighttime lights; spatial spillovers; Chile.

JEL codes: D74; O18; R11.

1 Introduction

Conflict is often measured where violence occurs, but its economic consequences may extend to towns and small cities that organize nearby rural activity. If rural conflict changes expectations about future safety, property-rights stability, or the contestability of nearby land, its consequences may appear not only as direct rural damage, but also as weaker formal activity in nearby urban

markets.

This paper asks whether nearby rural conflict is associated with weaker formal economic activity in exposed urban markets, and where any association is concentrated within urban areas. The empirical design therefore focuses on conflict outside the receiving urban market rather than on direct conflict within that market.

Two strands of prior work make rural-to-urban spillovers plausible. Evidence on violence, insecurity, and asset markets shows that perceived risk can affect vacancy, prices, and investment even where violence is not directly observed (Abadie and Dermisi, 2008; Guidolin and La Ferrara, 2010; Zussman et al., 2008; Besley and Mueller, 2012). Conflict-economics work further shows that violence can reshape firms, input use, market participation, and the composition of local activity rather than producing uniform output collapse (Camacho and Rodriguez, 2013; Del Prete et al., 2023; Galdo et al., 2020). Related work on land conflict and Indigenous property rights shows how historical dispossession, legal restrictions, and changes in the recognition of collective claims can shape both conflict incentives and investment decisions (Dippel et al., 2020; Castañeda Dower and Pfütze, 2020; Feir et al., 2023). What remains less understood is whether conflict that occurs mainly in rural areas reaches nearby urban markets, and whether that response is concentrated in particular areas within those markets.

Southern Chile is informative because a long-running territorial conflict unfolds largely in the rural hinterland, while nearby towns have formal property markets and unusually rich administrative records. The conflict follows the late-nineteenth-century incorporation of Mapuche territory, forced settlement, and subsequent policies that fragmented land rights and only partially repaired them through restitution (Jaimovich and Jordán, 2025b; Jaimovich and Toledo-Concha, 2026). Contemporary Mapuche mobilization centers on self-determination, land restitution, and territorial control rather than on conventional civil war or large-scale armed confrontation. Its events include attacks, coercive actions, protests, and land invasions, but most occur outside urban areas (Cayul et al., 2022).¹

Historically recognized claims, legal restrictions, and an active restitution regime can make territorial contestability more consequential for firms and households (Jaimovich and Jordán, 2025b; Jordán et al., 2025; Jaimovich and Toledo-Concha, 2026). The setting therefore permits the paper to trace conflict from the rural hinterland into formal outcomes in nearby urban markets and blocks. The institutional question extends beyond Chile. Colombia’s Decree Law 4633 provides for territorial reparation and restitution for Indigenous peoples affected by armed conflict; Canadian treaty and title claims and the tenure rules governing Native American reservation land show other ways historically grounded territorial rights continue to shape land use and local development (República de Colombia, 2011; Feir et al., 2023; Dippel et al., 2020).

This paper makes two contributions. First, it studies a rural-to-urban spillover margin in the economics of conflict. Existing work has studied the costs of violence for directly exposed firms,

¹In the study area, 82.5 percent of attacks, coercive actions, and land invasions fall outside the 2017 urban boundaries used in the analysis (Table 2).

households, regions, and assets (Camacho and Rodriguez, 2013; Del Prete et al., 2023; Besley and Mueller, 2012; Abadie and Dermisi, 2008). In southern Chile, the closest aggregate benchmark documents regional economic costs of violence (Larroulet Vignau and Sepúlveda, 2025). This paper moves inside the territory by asking whether rural conflict is associated with formal activity in towns and cities connected to it, even when those urban places are not themselves attacked.

Second, it separates changes in urban formal activity across urban markets from the spatial incidence of exposure within them. Aggregate estimates capture market-level responses, while block-level estimates show whether those responses are concentrated in particular parts of the urban area.

South-central Chile also makes the regional-development stake concrete. Rural households and firms rely on nearby towns for services, trade, formal business registration, and property transactions. If conflict weakens activity in those towns, it can burden the urban centers that connect the rural hinterland to wider markets. The estimates address that municipal link, not the broader causes of southern Chile’s long-run underdevelopment.

Empirically, the analysis follows urban markets and urban census blocks (*manzanas*) in Biobío, Araucanía, Los Ríos, and Los Lagos. This four-region study area is selected for substantive rather than nationally representative reasons. It covers the part of south-central Chile where historically documented Mapuche and Huilliche territorial occupation, titled-land legacies, contemporary restitution processes, and the recent geography of rural conflict most clearly overlap (Comisión Verdad Histórica y Nuevo Trato con los Pueblos Indígenas, 2008; Jaimovich and Jordán, 2025b; Jaimovich et al., 2025; Cayul et al., 2022). Appendix Figure A1 illustrates the historical territorial delineation that motivates this geography.

The empirical strategy links geocoded conflict events from the Mapuche Conflict Event and Data Archive (MACEDA) to urban markets and nearby urban blocks. The main treatment counts non-protest main events within 30 kilometers that occur outside the receiving urban market. These include rural events and events inside a different urban market. Annual conflict inside the receiving market enters separately as a concurrent co-exposure. SII enterprise counts and tax-reported sales measure aggregate formal activity, while nighttime lights, cadastral outcomes, and built-form data extend the evidence across other urban margins (Gibson et al., 2021; Huang et al., 2021; Martorano et al., 2024).

The main estimates show lower tax-reported sales and fewer enterprises in more exposed urban markets, conditional on own-market conflict. They do not show an equally broad contemporaneous change in lights, appraisals, or new built area, and block-level comparisons do not reveal a systematic within-market reallocation. Continuous timing profiles describe short-run sales associations and later movements in enterprises and some appraisal measures, but pre-exposure movement in several outcomes weakens a causal timing interpretation. The appendix complements the continuous profiles with staggered-DiD timing diagnostics around conflict spikes. One coding adapts the local-peak event rule in Barrett et al. (2022), while the threshold adaptations draw on the cumulative-exposure idea in Neri-Lainé (2026). These specifications provide a discrete view of

conflict salience alongside the continuous exposure measure.

The remainder of the paper proceeds as follows. Section 2 sets out the institutional history and legal framework that make nearby urban spillovers plausible. Section 3 positions the paper in the spillover, conflict, and Indigenous land-institutions literatures. Section 4 presents the data and empirical design. Section 5 reports the main estimates. Sections 6–7 assess alternative interpretations and identification limits, and the final section concludes.

2 Context and Background

2.1 Regional development backdrop

Southern Chile’s territorial conflict is concentrated in a region that has persistently lagged the rest of the country. Table 1 shows that La Araucanía ranked last nationally in GDP per capita in the 1992, 2002, and 2017 census-year snapshots. Relative to the Chilean average, it also trailed the neighboring southern macro-zones in each comparison.

Table 1: Regional GDP-per-capita snapshots in southern Chile

	1992	2002	2017
<i>Panel A. La Araucanía’s national GDP-per-capita rank</i>			
La Araucanía	13/13	13/13	16/16
<i>Panel B. Southern regions relative to Chile average (Chile = 100)</i>			
Chile average	100.0	100.0	100.0
Historical Biobío	76.6	78.3	69.6
La Araucanía	42.5	46.9	56.1
Historical Los Lagos	54.7	71.0	77.0
Los Ríos (2017 only)	—	—	65.5
Los Lagos (2017 only)	—	—	82.3

Panel A uses source-year regional geography: 13 regions in 1992 and 2002, and 16 in 2017. Panel B normalizes each region’s GDP per capita to the same-year Chile average after harmonizing Banco Central GDP and census population to identical boundaries. Historical Biobío combines Biobío and Ñuble where needed, and historical Los Lagos combines Los Ríos and Los Lagos; the split rows appear separately only in 2017.

The event data show a similarly concentrated geography. Excluding protests, La Araucanía accounts for 66.6% of Mapuche conflict events in the four-region study area, despite accounting for 25.7% of its 2017 population. These facts describe a persistent regional-development puzzle; they do not show that conflict caused La Araucanía’s relative underdevelopment. The possibility that conflict and weak development reinforce one another motivates the conflict-trap literature (Collier et al., 2003). This paper asks a more specific question: whether nearby rural conflict weakens formal activity in the urban markets that organize the surrounding hinterland.

2.2 Historical formation of claim salience

The Mapuche are the largest Indigenous group in Chile and have a long history of territorial autonomy and resistance to colonial and state incorporation. The Mapuche retained territorial autonomy through most of the colonial period and were incorporated into the Chilean state only after the military occupation of Araucanía in the late nineteenth century. Between 1884 and 1929, the state confined Mapuche communities to nearly 3,000 reservations, covering roughly 550,000 hectares and only a small share of ancestral territory (Jaimovich and Toledo-Concha, 2026; Jaimovich and Jordán, 2025b). These reservations were formalized through *Títulos de Merced* (TDM). In this paper, their importance is not that they describe current ownership. It is that they created a state-recognized geography of claims that later legal protection, restitution policy, and conflict continued to reference.

That original geography became increasingly detached from actual control over land. Allotment, sales, weak enforcement, and later deregulated transfers reduced Mapuche ownership within reservation boundaries over time (Jaimovich and Jordán, 2025b,a). For the empirical analysis, TDM proximity therefore measures claim salience rather than Indigenous ownership. A nearby block or urban market may be close to a historically recognized claim even when the surrounding land is currently held, used, or contested by different actors.

2.3 Agrarian reform and territorial restructuring

Chile’s agrarian reform began in 1962 and accelerated under Salvador Allende’s government between 1970 and 1973. It expropriated large estates, and Mapuche communities in the south recovered or gained access to part of that land through the reform. The process therefore reopened territorial claims beyond the boundaries of the historical reservations (Jaimovich and Toledo-Concha, 2026; Jaimovich and Jordán, 2025a).

The 1973 military coup reversed that trajectory. The dictatorship returned many expropriated estates to former owners or reassigned them to new private owners, while promoting a rural development path centered on subsidized forestry and large plantations. It also restructured Mapuche property itself. Decree Law 2,568 divided collectively held reservations into individual parcels, weakening the collective tenure system that had survived on much of the remaining Mapuche land (Jaimovich and Jordán, 2025a).

The reversal left a durable geography of grievance. Plots expropriated during the reform and later reversed or targeted by restitution are more likely to experience contemporary invasions and attacks (Jaimovich and Toledo-Concha, 2026). The land conflict therefore extends beyond the original reservation boundaries, linking contemporary events to both the failed reform and the property structure created after 1973.

2.4 The 1993 Indigenous Law: restrictions and enforcement

Democratization created a new legal framework, but not a settled property-rights environment. The 1993 Indigenous Law recognized Indigenous individuals and communities as legal holders of land and as parties to restitution. Article 12 defines Indigenous land through legal history, including *Títulos de Merced*, long-standing community occupation, later judicial recognition, and land transferred by the state (Gobierno de Chile, 1993). Protection can therefore attach to land because of how it entered the legal record, even when current ownership, registration, and enforcement have changed over time.

Article 13 does not impose one uniform restriction on all Indigenous land. It bars alienation, attachment, encumbrance, and acquisition by prescription, except for transfers among Indigenous persons or communities of the same ethnic group. CONADI may authorize an encumbrance, but not over the Indigenous family home or land needed for subsistence. Community land cannot be leased, lent, or ceded to third parties, while individually held Indigenous land may be leased or granted for use for up to five years. Individual owners may also exchange land for non-Indigenous land of similar commercial value with CONADI approval; the land received in exchange then becomes Indigenous land. Contracts that breach these rules are absolutely null (Gobierno de Chile, 1993).

The law’s protection also depended on whether those restrictions could be recognized and enforced. Article 15 created the Public Registry of Indigenous Territories (PRIT), where registration evidences Indigenous-land status (Gobierno de Chile, 1993). In property records from the Temuco area, Jordán et al. (2025) show that properties listed in PRIT have lower post-inheritance sales and smaller losses of Mapuche ownership, while registry coverage declined over time. These patterns make administrative enforcement a central issue in the practical protection of legally defined Indigenous land.

2.5 Restitution and claim salience

Restitution policy made that geography active again. Through the Fund for Indigenous Land and Water, the Chilean state has returned more than 200,000 hectares to Mapuche communities since the mid-1990s (Jaimovich et al., 2025). CONADI administers that process and therefore links historical claims to current state action. Restitution can resolve some claims, but it can also make surrounding claims more visible and more actionable (Jaimovich et al., 2025; Jaimovich and Toledo-Concha, 2026).

Conflict events are predominantly rural, but they occur where claims, restrictions, and restitution prospects remain legally salient. Firms and households in nearby towns may therefore respond not only to physical insecurity, but also to uncertainty about the future use of land and the contestability of the surrounding territory.

3 Literature Review

3.1 Conflict spillovers, valuation, and nearby insecurity

The economic incidence of conflict need not coincide with the event location. Alfano and Cornelissen (2026) show that conflict along transport routes in Somalia raises food prices in distant markets, even after accounting for local violence. Their mechanism is disrupted market access rather than territorial conflict, but the design lesson is directly relevant: events in rural space can affect economic outcomes in places that are not directly attacked.

Valuation evidence shows why that transmission can also operate through expectations. Besley and Mueller (2012) find that house prices in Northern Ireland reflected violence and the expected peace dividend, with spillovers toward adjacent areas. Abadie and Dermisi (2008) show that, without a direct attack on Chicago, vacancy rose more around landmark buildings after 9/11 as perceived terrorism risk increased. Financial markets likewise react quickly to conflict escalation and de-escalation because they price expected costs as well as realized damage (Zussman et al., 2008; Guidolin and La Ferrara, 2010). Buonanno et al. (2013) provides related evidence that perceived insecurity can affect housing values through expectations.

These studies establish that insecurity can alter economic decisions outside the place directly hit, but they focus on transport prices, house values, office vacancy, or national financial assets. They do not examine whether a predominantly rural self-determination conflict changes formal activity, construction, assessed property stock, and within-market spatial incidence in nearby secondary urban markets.

3.2 Conflict, firms, and spatial reallocation

Recent reviews emphasize that conflict can affect investment, migration, and local development across a wide range of settings (Di Maio and Sunde, 2025; Vesco et al., 2025). At the firm level, Camacho and Rodriguez (2013) show that armed conflict increases plant exit in Colombia, while Del Prete et al. (2023) document reductions in sales, employment, and input use and clarify mechanisms related to input disruption and market competition. Related evidence shows that conflict can also reallocate activity across margins that aggregate proxies do not separate well, which is one reason to distinguish administrative measures of formal activity and assessed stock from broader proxies of local activity (Galdo et al., 2020). More generally, the investment literature shows that uncertainty can delay irreversible development decisions, especially in real estate (Bulan et al., 2009).

Most firm-level evidence studies direct exposure or administrative units that already contain the conflict. That is not the same question as whether nearby rural violence reaches urban places that are not themselves event locations. In the Chilean setting, Larroulet Vignau and Sepúlveda (2025) provide the closest region-level benchmark by documenting sizable GDP losses from violence in the south. The unresolved issue is the transmission margin inside the territory: whether nearby rural conflict is reflected in urban formal activity, construction, assessed stock, and within-market

spatial patterns.

3.3 Land institutions, territorial conflict, and competing mechanisms

Land institutions can determine where a nearby conflict shock becomes economically salient. Constraints on transfer, collateralization, and enforcement can affect investment through delayed transactions, thinner markets, lower collateral value, and weaker incentives to make long-lived investments (Field, 2007; Dippel et al., 2020). Clearer or more secure rights can also change conflict incentives themselves (Castañeda Dower and Pfutze, 2020). The economic salience of a claim may therefore depend on later legal and political changes that affect enforcement, bargaining power, and the expected durability of rights (Feir et al., 2023).

The Mapuche case belongs to a broader class of Indigenous self-determination conflicts in which territorial claims are central to contention (Cayul et al., 2022). Recent work on southern Chile shows how forced settlement, allotment, and later market pressures shaped the geography of historically recognized claims (Jaimovich and Jordán, 2025b,a). It also shows that restitution and the legacy of reform and counter-reform help structure where contemporary contestation persists (Jaimovich et al., 2025; Jaimovich and Toledo-Concha, 2026). In property records from the Temuco area, Jordán et al. (2025) find lower post-inheritance sales and smaller losses of Mapuche ownership for properties listed in the Public Registry of Indigenous Territories, while registry coverage declined over time.

This evidence motivates historically recognized claims as a source of heterogeneity in the effects of nearby conflict exposure. A stronger response in claim-salient places would be consistent with property-rights risk, but it would not by itself identify that channel.

A second interpretation concerns ethnic stigma and statistical discrimination. Christensen and Timmins (2023) show that discriminatory beliefs and constraints in housing search can distort location choices and willingness to pay even when physical quality is unchanged. In this setting, local Mapuche composition may shape how nearby insecurity is interpreted. Any resulting heterogeneity is evidence consistent with a stigma channel, not a clean test that separates stigma from claims, local politics, or state presence.

The remaining question is whether rural territorial conflict reaches nearby urban markets, and whether its spatial incidence varies more closely with historically recognized claims than with broad ethnic composition.

4 Data & Empirical Strategy

4.1 Data and setting

The empirical analysis uses two linked annual panels for Biobío, Araucanía, Los Ríos, and Los Lagos. The first is an urban-market-year panel, where each market is the fixed commune-urban

footprint formed by the 2017 urban *manzanas* within that commune.² This panel is the main layer for aggregate market-level associations. The second is an urban *manzana*-year panel for within-market spatial analysis. The resulting target universe contains 34,922 urban *manzanas* in 104 communes.

The conflict exposure data, census controls, and mechanism variables use this fixed geography consistently, so the unit of observation remains stable even when coverage differs across sources. SII business outcomes measure activity at the commune level and are attached to the corresponding fixed commune-urban footprint in the aggregate panel. Nighttime lights, cadastre, and built-form products are measured on fixed urban areas or *manzanas*. Conflict exposure comes from geocoded events in the Mapuche Conflict Event and Data Archive (MACEDA) (Cayul et al., 2022). The event panel used in the analysis records event type and location from 1990–2021. The main exposure family includes attacks, coercive actions, and land invasions; protests are retained as a separate event family rather than as part of the baseline treatment.

The four-region corridor is selected for substantive reasons. It covers the part of southern Chile where historically documented Mapuche and Huilliche territorial occupation, historically recognized claims, contemporary restitution processes, and territorial conflict exposure most clearly overlap. The sample therefore provides a comparable target area for studying whether rural territorial conflict spills into nearby urban activity and land-use margins. Figure 1 displays the study setting in Chile and the concentration of main conflict events.

Table 2 describes the event geography used to construct the treatment. The baseline family consists of MACEDA event types 1, 3, and 4—attacks, coercive actions, and land invasions. Section 4.3 defines the spatial rule that maps these main events into nearby exposure.

Table 2: Composition and regional concentration of main Mapuche conflict events

Event family	Events	Rural MACEDA	Share	Outside 2017 urban mask	Share
All MACEDA events	3,730	1,143	30.6%	2,672	71.6%
Main event family: attacks, coercion, land invasions	2,775	497	17.9%	2,288	82.5%
Protests	955	646	67.6%	384	40.2%
<i>Panel B. Regional concentration of main events</i>					
Region	Main events	Event share	2017 population	Population share	
Biobio	752	27.1%	1,556,805	41.8%	
La Araucanía	1,847	66.6%	957,224	25.7%	
Los Lagos	76	2.7%	828,708	22.2%	
Los Ríos	100	3.6%	384,837	10.3%	
<i>Panel C. La Araucanía share of cumulative main events</i>					
Window	Main events	La Araucanía events	Share		
1990–1992	40	30	75.0%		
1990–2002	589	401	68.1%		
1990–2017	1,990	1,371	68.9%		
1990–2021	2,775	1,847	66.6%		

Notes: The table uses MACEDA events in the four-region study corridor. Main events are attacks, coercive actions, and land invasions; protests are excluded from the baseline treatment.

Figure 2 complements the composition table by showing the time path of main Mapuche conflict events and their regional concentration within the corridor.

²The fixed geography uses the 2017 INE manzana and urban-area GIS layer because it is the oldest available source that provides the required block-level geometry for the study corridor.

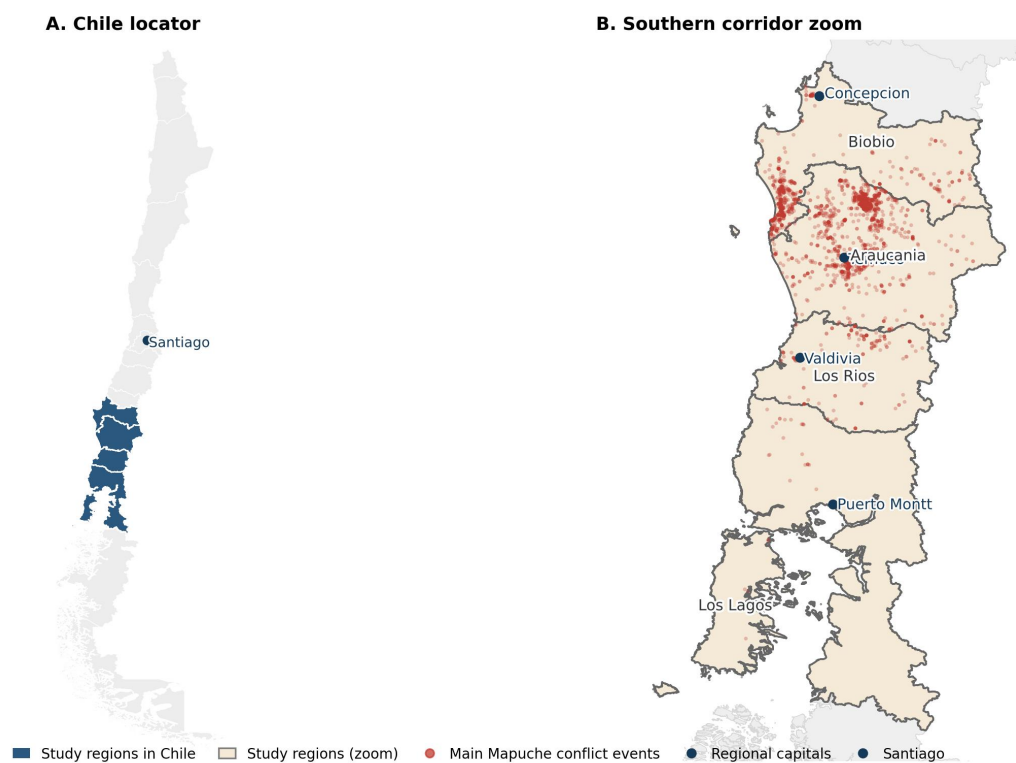


Figure 1: Study corridor in southern Chile. Panel A locates the four study regions—Biobío, Araucanía, Los Ríos, and Los Lagos—within Chile and marks Santiago for orientation. Panel B zooms into the southern corridor and displays main Mapuche conflict events.

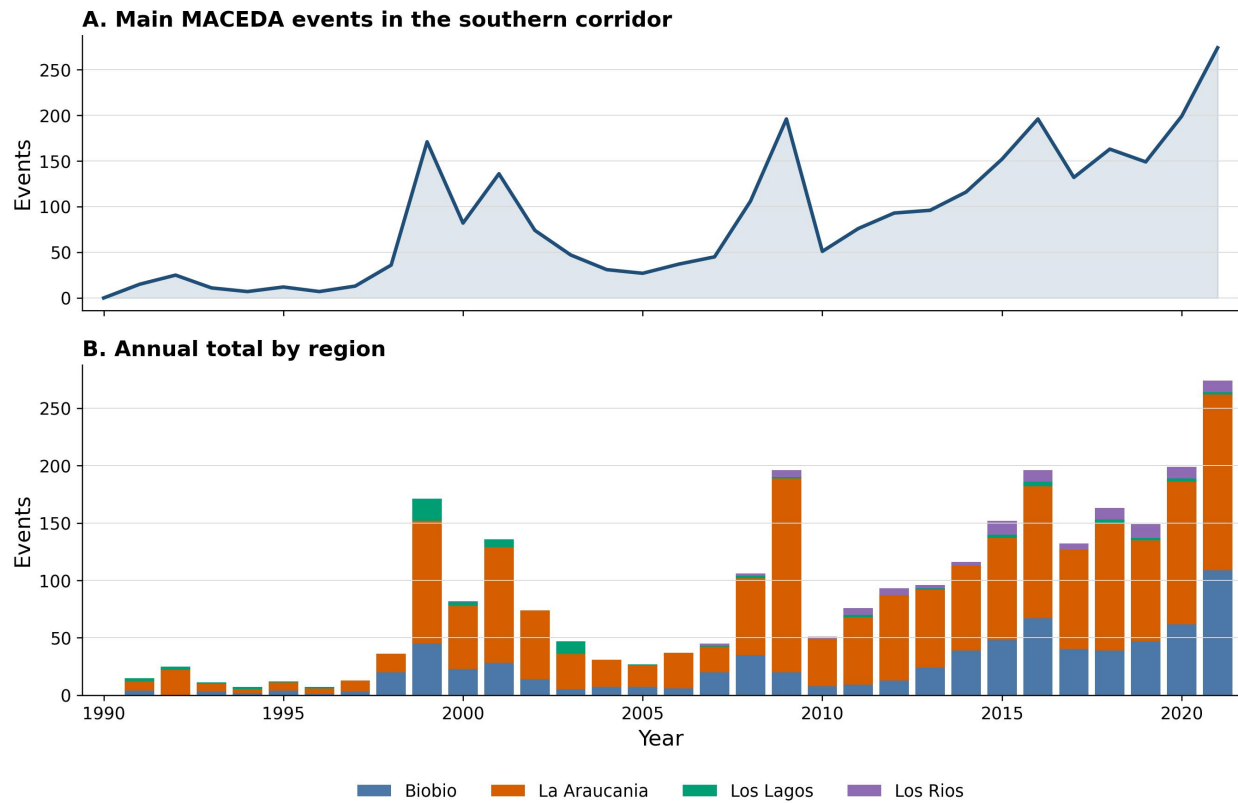


Figure 2: Main Mapuche conflict events in the four-region corridor. Panel A reports the annual count of attacks, coercive actions, and land invasions in the four-region corridor. Panel B reports annual main event counts separately for Biobío, Araucanía, Los Ríos, and Los Lagos.

4.2 Outcome measurement

The outcomes track four distinct margins: commune-level formal activity, the assessed and physical property stock, nighttime luminosity, and new built area. Administrative records measure the first two margins directly, while satellite products extend the temporal and spatial coverage of local activity and construction. SII enterprise counts and tax-reported sales measure formal activity registered in the commune containing each urban market. Because firms are assigned by tax address, these outcomes cannot isolate activity inside the fixed urban footprint. The aggregate estimates are therefore not direct measures of urban private investment. Appendix Table A5 separately reports agriculture and forestry, all available SII *rubro* records, and selected urban- and firm-facing sectors under the preferred specification. Agriculture and forestry come from the more detailed SII subrubro source; fishing and aquaculture are excluded. The table does not solve the tax-address problem, but it shows whether the aggregate pattern is confined to rural primary activity or also appears in the reported business records more broadly.

The SII cadastre measures the assessed and physical property stock. Fiscal appraisal is the tax-base assessment, not a transaction price, and changes reflect mass revaluations and indexation as well as changes in the registered stock (Servicio de Impuestos Internos, 2006, 2026a,b). Parcel counts, built area, and destination shares describe the composition of that stock. Because owners, municipalities, and public agencies all supply cadastral updates (Servicio de Impuestos Internos, 2025), annual changes may combine physical adjustment with administrative updating.

Satellite measures cover margins that the administrative records do not. Harmonized DMSP–VIIRS data measure aggregate luminosity, while annual VIIRS data provide finer spatial evidence from 2013 onward. I exclude the nominal 2012 VIIRS composite because it combines April–December 2012 with January–March 2013; assigning it to 2012 would match part of the light measure to conflict exposure from the wrong calendar year. The available data do not contain the monthly rasters needed to construct an April–December 2012 sensitivity. At the *manzana* level, VIIRS measures local luminosity rather than block output because most blocks cover only a fraction of a 500-meter pixel. WSF Evolution measures new built-up area through 2015, not sales, prices, or productivity.

Transformations follow the support of each outcome. Positive totals and enterprise counts enter in logs, destination shares in percentage points, and zero-inclusive VIIRS radiance and WSF construction in inverse hyperbolic sines. Appendix Tables A1 and A2 report coverage and sample support; the appendix also reports prespecified alternative scales.

4.3 Measuring nearby conflict exposure

Baseline exposure is the log of one plus the annual number of attacks, coercive actions, and land invasions occurring outside the receiving urban market and within 30 kilometers of the focal market or *manzana*. It combines events outside every corrected urban system with events inside another urban market. Main events inside the receiving market form a separate annual co-exposure. Protests are excluded. I form each urban system from the union of its fixed 2017 urban *manzanas*, close

streets and small gaps within 25 meters, and retain the external shells of components that remain distinct. The resulting edge is a fixed measurement boundary; it does not imply that every location had the same urban status throughout 1990–2021. Appendix Figure A8 illustrates the rule around Temuco and Cañete.

The baseline event family combines attacks, coercive actions, and land invasions. These events differ in physical damage, but each records coercive or violent territorial contestation. Their annual count measures the intensity of nearby insecurity rather than a common amount of damage.

Let $N_{mt}^{\text{out}}(r)$ denote the number of external main-conflict events within r kilometers of market m in year t , and let $N_{imt}^{\text{out}}(r)$ denote the corresponding count for *manzana* i in market m . The benchmark exposure indices are

$$E_{mt}^{\text{out}} = \log[1 + N_{mt}^{\text{out}}(30)] , \quad E_{imt}^{\text{out}} = \log[1 + N_{imt}^{\text{out}}(30)] . \quad (1)$$

The transformation reduces the influence of extreme event years; a one-unit change is a change in log one plus the event count, not one additional event. I define E_{mt}^{in} from all main events inside the receiving urban market in year t , with no distance cutoff. It is a competing urban co-exposure, not part of the external treatment. At the *manzana* level, the annual market-wide inside measure is constant within a market-year and is absorbed by the market-year fixed effects. It is therefore not a separately estimated block-level control. I use 30 kilometers as a common summary window and report the 10–60 kilometer sensitivity separately; the radius is not an estimated cutoff or a value chosen from the coefficients.

4.4 Empirical design and estimands

Nearby conflict is recurrent and varies in intensity; a first event does not create a permanent treatment state. The baseline estimand is therefore the conditional slope associated with annual exposure intensity. The aggregate design compares the same market across years, whereas the *manzana* design compares blocks within the same market and year. Each market is the fixed urban area associated with one commune, so the market index also identifies the corresponding administrative commune.³

4.4.1 Aggregate market-level design

Let Y_{mt} denote an outcome for urban market m in year t . The aggregate specification is

$$Y_{mt} = \beta_A E_{mt}^{\text{out}} + \mu_m + \lambda_t + \tau_t (X_m^0)' \theta + \varepsilon_{mt}, \quad (2)$$

³All regressions are unweighted. Aggregate markets with longer outcome support and more residualized exposure variation contribute more to the projection; at the block level, markets with more observed *manzanas* contribute more observations. Neither coefficient is a population-, output-, or market-equal-weighted average. A sensitivity that equalizes total market weight leaves the aggregate estimates essentially unchanged and produces no block-level sign reversal.

where μ_m and λ_t are market and year fixed effects, respectively, and τ_t is a linear calendar-time index measured in decades. The vector X_m^0 contains predetermined baseline conditions. The specification ladder begins with fixed effects alone, then adds baseline trends and a separate trend for administrative-center status. Standard errors are clustered by market.

The external-exposure specification estimates the conditional association between market outcomes and geographically external conflict intensity. A joint model adds E_{mt}^{in} , so β_A becomes the association with external exposure conditional on contemporaneous inside-urban conflict. The inside measure is a competing co-exposure rather than a predetermined control, and the joint model does not mechanically isolate an exogenous rural shock.

A causal interpretation requires changes in external exposure to be unrelated to unobserved market-specific outcome shocks after conditioning on the fixed effects and baseline trends. The trend interactions allow markets with different inherited conflict, economic scale, forestry exposure, road access, claim salience, and administrative status to follow different linear paths. They do not absorb contemporaneous shocks specific to one market. I exclude contemporaneous economic and land-use controls because they may respond to conflict.

4.4.2 Within-market manzana design

The block-level specification compares differential spatial incidence within local markets:

$$Y_{imt} = \beta_M E_{imt}^{\text{out}} + \gamma_i + \delta_{mt} + \tau_t(X_i^0)' \phi + \varepsilon_{imt}, \quad (3)$$

where γ_i are *manzana* fixed effects, δ_{mt} are market-year fixed effects, and X_i^0 contains predetermined block characteristics. Standard errors are clustered by market.

Manzana fixed effects remove persistent differences across blocks, while market-year fixed effects absorb business conditions, policies, and security shocks common to the market. The coefficient β_M therefore uses changes in relative exposure across blocks in the same market-year. Any market-wide response is absorbed by δ_{mt} , so this specification cannot estimate aggregate market loss. It supports a reallocation interpretation only if the estimates reveal a stable offsetting spatial pattern.

The exposure file retains direct inside-block assignments and same-market spillovers for diagnostic work. They do not enter the preferred specification: the block design instead identifies the association between external exposure and outcomes from relative exposure across blocks within the same market-year.

The outcome panels begin at different points in the conflict history. Conflict is observed from 1990 onward, while formal sales begin in 2005, fiscal appraisal mainly in 2010, preferred calendar-year VIIRS in 2013, and WSF near the start of the MACEDA window. Trend specifications therefore use pre-window conflict histories only when such a window exists: pre-2005 exposure for SII business outcomes and pre-2010 or pre-2012 exposure for later administrative or satellite outcomes. Outcomes beginning at or near the start of MACEDA, such as historical nighttime lights and WSF, do not include a pre-window conflict trend by default. Baseline trend controls

also include predetermined census and geographic variables such as population scale, density, rural share, Mapuche composition, forestry development, and capital status. In the aggregate models, road access is measured around the urban market and land-rights salience by the share of the market overlapping TDM polygons. In the *manzana* models, road access remains block-level, while TDM and CONADI proximity are reserved for the heterogeneity analysis rather than included as baseline controls.

4.5 Timing diagnostics: distributed lags and salient shocks

As a timing check, I ask whether outcomes move before high-conflict years and whether adjustment appears immediately or with lags. The first diagnostic is a distributed lead-lag version of the continuous exposure model:

$$Y_{mt} = \sum_{k=-3}^L \beta_k E_{m,t-k}^{\text{out}} + \mu_m + \lambda_t + \tau_t (X_m^0)' \theta + \varepsilon_{mt}, \quad (4)$$

where $k = -3, -2, -1$ indexes future exposure, $k = 0$ current exposure, and positive values lagged exposure. The maximum lag L is eight or ten years, depending on outcome support. I report each future-exposure coefficient and a joint Wald test of the future terms. These coefficients show whether the outcome was already moving before high-exposure years, while the current and lagged terms describe the timing of adjustment. Because exposure is serially correlated, the future terms are diagnostics for prior movement rather than a stand-alone test of identification.

The second diagnostic studies unusually large escalations in individual markets. It uses a local-peak rule that adapts the salience logic of Barrett et al. (2022): a spike is a local peak with at least two main events within 30 kilometers and an extreme annual count relative to the market’s history or recent baseline. I also report a cumulative threshold adaptation motivated by the exposure accumulation considered in Neri-Lainé (2026), alongside a recurrent annual threshold sensitivity created for this paper. These are alternative ways to summarize conflict salience, not replications of the source studies.

For each coding, Callaway–Sant’Anna group-time effects recode the first qualifying year as an absorbing onset and compare cohorts with not-yet-treated markets (Callaway and Sant’Anna, 2021). A complementary de Chaisemartin–D’Haultfoeuille representation retains the annual binary path, so markets may return to zero or experience another spike. The resulting profiles provide a discrete timing description alongside the continuous lead-lag model.

4.6 Heterogeneity and mechanism interpretation

The property-rights and stigma hypotheses motivate exploratory heterogeneity by TDM salience, predetermined Mapuche presence, and event composition. These characteristics are not randomly assigned mediators, so their interactions describe differences across fixed geographies rather than causal mechanisms. The primary interaction model enters nearby TDM share and the 1992 Ma-

puche population share jointly, together with the matching interactions for annual own-market conflict. It uses the same aggregate fixed-effects specification as the preferred main estimates, market-clustered inference, and a prespecified within-family multiplicity adjustment.

CONADI polygon interactions are excluded because the available recent snapshots do not identify the land at risk before each outcome year. A separate replication diagnostic attaches dated Article 20(a) and 20(b) purchase records to aggregate markets through commune identifiers, but it measures a commune-wide pre-period purchase history rather than land close to the urban footprint. That history is itself selected by claims and institutional action, so it is not used as a causal treatment or a paper-facing mechanism test. Fatality and property-damage splits are also excluded because nonmissing support is too sparse.

5 Main Results

5.1 Aggregate urban-market evidence

Table 3 reports conditional associations for formal activity, nighttime lights, assessed property values, and new built area. Panel A adds controls sequentially for the 30-kilometer summary measure. Panels B and C show the cumulative and exact-band versions of the same broad external exposure through 60 kilometers.

Conditional on annual own-market internal conflict, sales and enterprise counts are lower in more exposed markets. Moving from the 25th to the 75th percentile of positive external exposure corresponds to a 7.24 percent decline in sales. These estimates describe municipal formal activity around fixed urban markets, not activity confined to their urban footprints.

Urban nighttime lights, fiscal appraisal, and new built area do not show a corresponding contemporaneous decline. The formal-activity association is not a uniform change across every aggregate urban measure.

Appendix Figures A2 and A3 report correlations between SII sales and nighttime lights. They are strongly correlated in levels, but the relationship becomes weak after removing market and year fixed effects. The residual correlation is about 0.10 for urban lights and about 0.16 for municipal or non-urban lights. Nighttime lights therefore do not closely measure the within-market formal activity variation used in the SII regressions.

The estimates do not identify a natural 30-kilometer boundary. Sales remains negative at 30 and 40 kilometers under the Bonferroni intervals for the cumulative profile, but the other aggregate outcomes do not show the same pattern. The estimates for the outer 30–60 kilometer bands do not jointly reject zero at the five-percent level. I use 30 kilometers as a common reporting window, not as a literal range of impact.

5.2 Spatial incidence within urban markets

Table 4 uses *manzana*-year outcomes to compare more and less exposed blocks within the same local market and year.

Table 3: Aggregate urban-market estimates

	Formal activity		Activity proxy	Assessed stock		Built-form flow
	(1)	(2)	(3)	(4)	(5)	(6)
	SII sales	Enter- prises	Urban NTL	Fiscal appraisal sum	Fiscal appraisal mean	WSF new
<i>Panel A. Common 30 km broad external exposure (A)</i>						
Market FE + year FE	-0.063*** (0.021)	-0.024*** (0.009)	-0.008 (0.011)	-0.012 (0.023)	-0.005 (0.022)	-0.060* (0.035)
+ baseline trends	-0.047*** (0.015)	-0.012** (0.005)	0.008 (0.009)	-0.000 (0.016)	-0.002 (0.015)	-0.045 (0.033)
+ capital trend	-0.049*** (0.015)	-0.012** (0.005)	0.009 (0.009)	0.002 (0.016)	-0.001 (0.015)	-0.045 (0.033)
+ all own-market I (no radius)	-0.049*** (0.015)	-0.012** (0.005)	0.007 (0.009)	0.002 (0.016)	-0.000 (0.015)	-0.037 (0.033)
<i>Panel B. Cumulative broad external exposure (A), 10–60 km; annual I has no radius cutoff</i>						
Within 10 km	-0.034 (0.024)	-0.010 (0.007)	0.014 (0.014)	0.020 (0.015)	0.017 (0.014)	0.042 (0.044)
Within 20 km	-0.035** (0.015)	-0.009 (0.005)	0.017* (0.010)	0.015 (0.014)	0.013 (0.013)	-0.025 (0.032)
Within 30 km	-0.049*** (0.015)	-0.012** (0.005)	0.007 (0.009)	0.002 (0.016)	-0.000 (0.015)	-0.037 (0.033)
Within 40 km	-0.054*** (0.018)	-0.012** (0.005)	0.005 (0.009)	-0.001 (0.016)	0.002 (0.015)	-0.053* (0.031)
Within 50 km	-0.044** (0.018)	-0.009* (0.005)	0.002 (0.009)	0.004 (0.015)	0.010 (0.014)	-0.028 (0.033)
Within 60 km	-0.043** (0.018)	-0.008 (0.006)	0.002 (0.008)	0.005 (0.015)	0.006 (0.014)	-0.050 (0.033)
<i>Panel C. Joint non-overlapping broad external bands (A), 0–60 km; annual I has no radius cutoff</i>						
0–10 km	-0.029 (0.023)	-0.008 (0.007)	0.008 (0.014)	0.018 (0.015)	0.016 (0.014)	0.066 (0.045)
10–20 km	-0.003 (0.019)	-0.005 (0.006)	0.014 (0.014)	0.007 (0.016)	0.006 (0.015)	-0.048 (0.036)
20–30 km	-0.035** (0.017)	-0.008 (0.005)	-0.005 (0.010)	-0.005 (0.017)	-0.008 (0.016)	-0.005 (0.045)
30–40 km	-0.013 (0.017)	-0.004 (0.006)	0.008 (0.011)	-0.016 (0.018)	-0.012 (0.017)	-0.057* (0.029)
40–50 km	0.007 (0.018)	0.005 (0.007)	0.002 (0.010)	-0.002 (0.016)	0.003 (0.015)	0.054 (0.034)
50–60 km	0.015 (0.013)	0.000 (0.005)	0.010 (0.010)	-0.031** (0.012)	-0.028** (0.012)	-0.036 (0.034)
Observations	1,768	1,768	3,060	1,248	1,248	2,704
Markets	104	104	104	104	104	104
Window	2005–2021	2005–2021	1992–2021	2010–2021	2010–2021	1990–2015

Notes: Each cell reports the coefficient on broad external conflict within the stated distance, with urban-market-clustered standard errors below. Broad external exposure includes events outside every corrected urban system and events inside another urban market; protests are excluded. Panel A's final row and all radius specifications condition on annual main events inside the receiving market, with no distance cutoff. All models include market and year fixed effects, predetermined baseline-condition trends, and a capital-status trend. Positive SII, nighttime-light, and cadastre totals enter in logs; WSF new built area enters in inverse hyperbolic sine. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Manzana-level spatial incidence estimates

	VIIRS mean	Fiscal sum	Fiscal mean	Commerce share	WSF new
<i>Panel A. Broad external A cumulative-radius profile; market-year FE absorb annual own-market I</i>					
Within 10 km	-0.020 (0.012)	0.016 (0.025)	0.009 (0.016)	-0.014 (0.022)	0.002 (0.001)
Within 20 km	0.012 (0.019)	0.023 (0.020)	0.012 (0.015)	-0.004 (0.022)	-0.002 (0.003)
Within 30 km (common summary)	-0.007 (0.021)	0.034 (0.022)	0.038** (0.019)	0.028 (0.032)	0.002 (0.003)
Within 40 km	-0.004 (0.016)	-0.001 (0.045)	0.031 (0.025)	-0.058 (0.048)	0.002 (0.004)
Within 50 km	0.012 (0.025)	0.056** (0.023)	0.037 (0.025)	-0.039 (0.050)	0.001 (0.004)
Within 60 km	0.031 (0.033)	-0.003 (0.024)	0.003 (0.024)	0.002 (0.056)	0.007*** (0.002)
<i>Panel B. Joint exact broad external A bands; market-year FE absorb annual own-market I</i>					
0–10 km	-0.015 (0.016)	0.033 (0.031)	0.025 (0.019)	-0.037 (0.027)	0.003 (0.002)
10–20 km	0.009 (0.016)	0.027 (0.021)	0.024* (0.014)	-0.030 (0.024)	0.002 (0.004)
20–30 km	-0.004 (0.013)	0.007 (0.019)	0.019 (0.015)	-0.023 (0.025)	0.003* (0.002)
30–40 km	-0.002 (0.013)	-0.002 (0.017)	0.008 (0.012)	-0.054** (0.027)	0.002 (0.002)
40–50 km	0.001 (0.013)	0.007 (0.011)	0.005 (0.011)	-0.022 (0.030)	0.000 (0.002)
50–60 km	0.015 (0.014)	-0.022* (0.011)	-0.014 (0.012)	-0.020 (0.027)	0.004** (0.002)
Observations	314,298	393,800	393,800	331,991	209,532
Manzana fixed effects	Yes	Yes	Yes	Yes	Yes
Commune-year fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: Each cell reports the coefficient on broad external exposure, using minimum event-to-manzana-polygon distance, with commune-clustered standard errors below. Panel A uses the 30-kilometer summary measure; Panel B estimates the six exact 10-kilometer bands jointly. All models include manzana and market-year fixed effects and predetermined baseline trends. Annual own-market events are constant within a market-year and are absorbed by the market-year effects rather than separately estimated. The commerce outcome is the percentage share of cadastre properties assigned to commercial use; VIIRS and WSF use inverse-hyperbolic-sine transformations. Protests are excluded. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

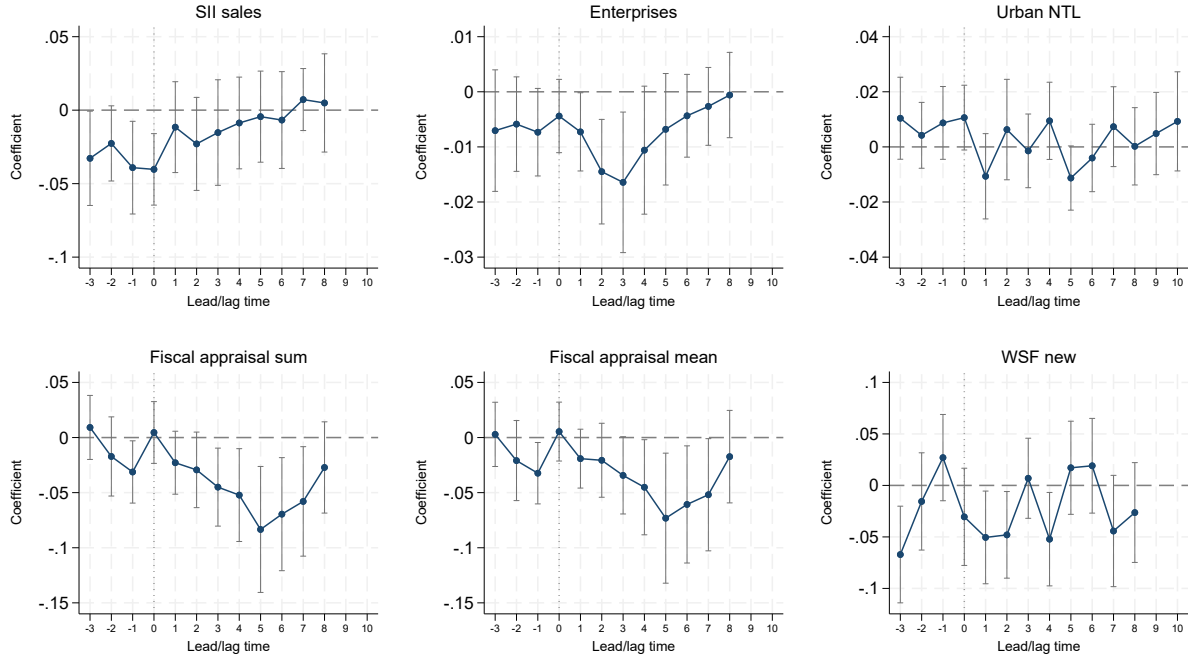
After market-year fixed effects are included, the preferred within-market estimates do not show a consistent pattern. VIIRS lights, fiscal-appraisal sums, new built area, and commercial use are imprecise. Fiscal-appraisal mean is positive, but one coefficient does not establish a stable within-market response.

The aggregate evidence points to fewer formal enterprises in exposed urban markets, but more exposed blocks do not systematically lose lights, assessed value, new built area, or commercial use relative to other blocks in the same market-year. The *manzana* design therefore does not show a clear reallocation of losses within urban markets.

The distance-band estimates also vary without a stable within-market gradient.

5.3 Timing of aggregate and within-market responses

Figure 3 and Table 5 examine whether aggregate outcomes move before or after changes in nearby external conflict exposure.



Broad external A within 30 km, conditional on annual own-market I with no distance cutoff. Negative k values are future-exposure diagnostics, not a sufficient parallel-trends test.

Figure 3: Dynamic aggregate response to nearby external conflict exposure. The figure reports coefficients from panel fixed-effects lead-lag specifications using log broad-external exposure within 30 kilometers, conditional on annual own-market conflict with no distance cutoff. Negative values of k are future exposure terms, $k = 0$ is current exposure, and positive values are lagged exposure terms. Vertical bars are 95 percent confidence intervals clustered by urban market. All models use the prespecified outcome transformations.

The joint future-exposure test rejects at the 10 percent level for SII sales ($p = 0.083$) and at the 5 percent level for fiscal-appraisal totals ($p = 0.027$), fiscal-appraisal means ($p = 0.035$), and WSF

Table 5: Aggregate dynamic summary

	(1)	(2)	(3)	(4)	(5)	(6)
	SII sales	Enter- prises	Urban NTL	Fiscal appraisal sum	Fiscal appraisal mean	WSF new
Joint leads -3/-1	2.281* p=0.084	1.168 p=0.326	0.863 p=0.463	3.159** p=0.028	2.963** p=0.036	3.395** p=0.021
Average post 0-2	-0.025** (0.012)	-0.009*** (0.003)	0.002 (0.006)	-0.016 (0.013)	-0.011 (0.012)	-0.043*** (0.013)
Average post 3-5	-0.009 (0.013)	-0.011** (0.006)	-0.001 (0.004)	-0.060*** (0.022)	-0.051** (0.022)	-0.009 (0.012)
Average post 6-8	0.002 (0.011)	-0.003 (0.003)	0.001 (0.004)	-0.052** (0.023)	-0.043* (0.024)	-0.017 (0.013)
Observations	1,456	1,456	1,964	936	936	1,872
Markets	104	104	104	104	104	104
Window	2005–2018	2005–2018	2000–2018	2010–2018	2010–2018	1998–2015

Notes: The first row reports the joint test of the future-exposure coefficients for $k = -3$, $k = -2$, and $k = -1$. Remaining rows report covariance-aware average post-exposure coefficients over the indicated lag windows, with urban-market-clustered standard errors below. All specifications include urban-market and year fixed effects, baseline-condition trends, a capital-status trend, and annual own-market internal conflict with no distance cutoff.

new built area ($p = 0.020$). It does not reject for enterprises ($p = 0.334$) or urban nighttime lights ($p = 0.464$). The profiles therefore show pre-exposure movement in several outcome series rather than offering reassurance about timing. They are not parallel-trends tests: exposure is recurrent and serially correlated, and the coefficients come from a distributed projection rather than a single treatment transition.

The post-exposure profiles also differ across outcomes. Sales average -0.025 over years 0–2, while enterprises are most negative over years 3–5 (-0.011). Fiscal-appraisal totals and means decline over years 3–5, and WSF new built area declines over years 0–2. The profiles describe timing and persistence; they do not establish a common causal adjustment path or resolve the distance evidence.

At the block level, the static common-sample distance profiles remain the relevant evidence. They do not reveal a stable relative decline across the measured urban outcomes.

6 Exploratory Heterogeneity

Historical claim geography and baseline Mapuche presence may change how nearby conflict correlates with local outcomes. These are fixed or predetermined features of places, not randomly assigned mediators. The estimates therefore describe exploratory heterogeneity, not a property-rights or stigma mechanism.

Appendix Figure A19 reports the pre-specified aggregate interaction model. It enters the 10-kilometer nearby TDM share and the 1992 Mapuche population share jointly, as well as the corresponding interactions with annual own-market conflict. The specification otherwise follows the preferred aggregate model: market and year fixed effects, baseline-condition trends, the capital-

status trend, and market-clustered inference. The figure reports every interaction across the six outcomes, and Appendix Table A6 reports the complete wild-bootstrap and multiplicity family.

The point estimates do not support a clean moderator result. The TDM interaction for urban lights is negative before adjustment, but none of the twelve focal interactions has a Benjamini–Hochberg adjusted wild-bootstrap value below five percent. The same exercise does not show a systematic difference by 1992 Mapuche share. These patterns are useful limits on a mechanism interpretation: the aggregate association does not become a property-rights or stigma result simply because it is stronger in one historical geography.

The remaining packages stay outside the paper-facing evidence. Recent CONADI polygon snapshots cannot recover the land at risk before each outcome year. Commune-level purchase histories can be constructed from dated records, but they are selected institutional responses and do not locate purchases near the urban footprint. Initiator and event-type decompositions describe the realized conflict process rather than exogenous treatment margins. Fatalities are unknown for 98.8 percent of main events and property damage for 95.0 percent; those fields are excluded rather than recoded as zero.

7 Robustness and Identification Limits

The static SII associations could reflect shocks that coincide with nearby conflict. The paper cannot remove that concern with fixed effects and predetermined trends alone. The most useful checks therefore ask whether the pattern survives changes in distance, timing, and the composition of the tax-address outcomes.

The distance results do not point to a single spatial boundary. Sales remains negative at 30 and 40 kilometers under the Bonferroni intervals for the cumulative profile, while the other aggregate outcomes do not show the same pattern. The outer 30–60 kilometer bands do not jointly reject zero at the five-percent level, and the block-level profile lacks a stable gradient. These checks support the use of 30 kilometers as a common summary window, not as an estimated range of impact.

The continuous lead-lag profiles are timing diagnostics rather than a parallel-trends test. The joint future-exposure test rejects for sales at the 10 percent level and for fiscal-appraisal totals, fiscal-appraisal means, and new built area at the 5 percent level; it does not reject for enterprises or urban nighttime lights (Table 5). Pre-exposure movement in several outcome series means that these profiles do not support a timing-based exogeneity claim.

The appendix adds discrete timing representations around unusually high conflict years. It adapts a local-peak coding from the salience logic in Barrett et al. (2022) and cumulative and recurrent threshold adaptations motivated by Neri-Lainé (2026). C&S treats the first qualifying year as an absorbing onset, while de Chaisemartin–D’Haultfoeuille retains the annual spike path. These profiles describe alternative timing patterns; they do not create an external source of conflict variation or establish causal event-study effects.

The tax-address limitation is equally important. Appendix Table A5 separately reports agri-

culture, forestry, all SII *rubro* records, and the most relevant urban- and firm-facing sectors under the preferred specification. Agriculture and forestry come from SII subrubro categories; fishing and aquaculture are excluded. The all-rubro records remain negative for both sales and enterprise counts. Agriculture sales decline, while forestry is imprecise; transport and storage declines on both margins. The decomposition does not locate activity inside the fixed urban footprint, but it shows that the aggregate pattern is not a common movement across every rural primary sector.

Nighttime lights do not resolve the same measurement problem. Appendix Figures A2 and A3 show that tax-reported sales and lights are closely related in levels but only weakly related after removing market and year fixed effects. The aggregate formal-activity estimates should therefore be read as municipal tax-address associations around urban markets, not as direct estimates of production inside the urban polygon.

8 Conclusion

This paper studies whether territorial conflict outside a receiving urban market is associated with formal economic activity around that market in southern Chile. The main aggregate estimates show lower tax-reported sales and fewer registered enterprises where broad external conflict is more intense, conditional on annual conflict inside the receiving market. These are municipal tax-address associations, not estimates of activity confined to the fixed urban footprint.

Other outcomes limit how far the result can be read. Nighttime lights, fiscal appraisal, and new built area do not show a comparable contemporaneous decline. More exposed blocks within the same market-year also do not systematically lose lights, assessed value, new built area, or commercial use relative to less exposed blocks. The distance checks do not identify a sharp boundary, and the timing profiles show prior movement in several outcome series.

The interpretation remains bounded by measurement and design. SII outcomes are municipal, fiscal appraisal is an administrative assessed-stock record rather than a transaction-price series, and the fixed-effects design does not supply an external source of variation in conflict. Within those limits, the results are consistent with territorial conflict affecting economic activity beyond the event location, even though the broader urban evidence does not show a general contraction.

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A Historical Geography of the Study Corridor

Figure A1 places the four study regions within the historical Mapuche territorial setting.

A. Historical territorial plate



B. Contemporary corridor and 1992 Mapuche shares

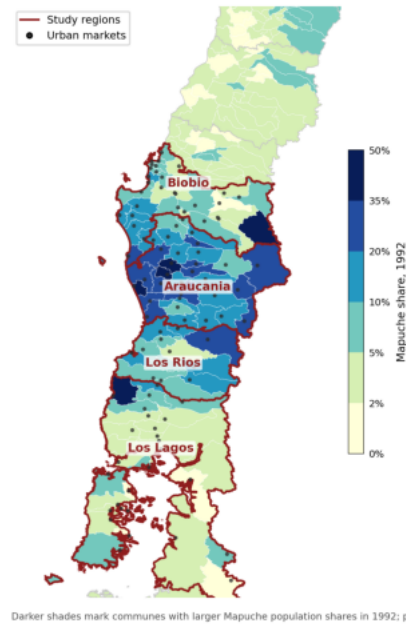


Figure A1: Historical Mapuche territory and the contemporary study corridor. Panel A reproduces a historical map of Mapuche territorial organization around 1540 from the Comisión Verdad Histórica y Nuevo Trato. Panel B shows that the contemporary study corridor lies in the south-central area where Mapuche population shares remained highest in the earliest modern census available in the MDP data.

B Data Sources, Time Coverage, and Sample Support

The empirical panels differ in start year and spatial support. The tables in this section record the data families, outcome windows, and geographic coverage behind the estimates reported in the main text.

Table A1: Summary of data families used in the paper

Role in paper	Family / outcome	Source	Time coverage	Geographic coverage / level
Treatment	Nearby external conflict exposure	MACEDA conflict archive	1990–2021	Corrected fixed urban-system edge $\times$ year; urban manzana $\times$ year
Pre-period exposure histories	Cumulative and lagged conflict exposure	Derived from the MACEDA exposure panel	1990–2009 summaries attached to later outcome samples	Manzana-level baseline variables
Aggregate formal activity	Enterprises and tax-reported sales	SII business statistics	Mainly 2005+	Commune $\times$ year; attached to the fixed commune-urban footprint
Assessed tax-base stock	Fiscal appraisal	Chilean tax cadastre (SII)	Mainly 2010+	Urban manzana
Physical cadastre composition	Registered properties and land-use composition; built area in later records	Chilean tax cadastre (SII)	Mainly 2010+; built area 2018+	Urban manzana
Aggregate luminosity proxy	Harmonized DMSP–VIIRS nighttime lights	Li–Zhou–Zhao–Zhao harmonized DN product	1992–2023; causal-aligned estimates through 2021	Fixed INE 2017 urban-market footprint $\times$ year
Within-market luminosity proxy	VIIRS nighttime lights	EOG VIIRS annual composites	2013–2021 preferred calendar-year sample	Urban manzana $\times$ year; exact fractional-pixel extraction
Historical lights robustness	DMSP nighttime lights	DMSP-OLS stable lights	1992–2013	Distrito $\times$ year
Canonical controls	Historical and fixed census controls	MDP, Redatam, and Censo 2017	1992 / 2002 / 2017 snapshots	Comuna-static repeated to manzana plus manzana-static baseline
Predetermined confounder layer	Road access, forestry-industry geography, and private plantation exposure where feasible	Road-network sources, forestry-industry layers, and plantation layers where available	Mainly static or pre-period baseline	Manzana-static or buffer-based baseline geography
Mechanism variables	Land-rights salience and restitution activity	Historical claim and CONADI polygon layers	Static	Manzana-static
Built-form extension	Annual newly built area	WSF Evolution first-built year	1985–2015	Fixed urban footprint or historical-support zone $\times$ year

The table reports the role, coverage, and interpretation of each data family in the paper.

SII business outcomes measure commune-level formal activity. Fiscal appraisal is an assessed tax-base stock, while the physical cadastre describes registered property and built-stock composition.

The aggregate series is a published harmonized DMSP–VIIRS product on a common DN scale. Native VIIRS and native DMSP remain separate in finer-scale analyses.

Predetermined geography and census variables enter through baseline trends in the augmented specifications.

Table A2: Sample support and time coverage by data family

Data family	Unit	Window	Obs.	Units	Comm. Role / coverage note
SII formal activity	Urban market-year	2005–2021	1,768	104 markets	104 Sales, enterprises, worker and payroll margins
Harmonized NTL	Urban market-year	1992–2021	3,060	104 markets	104 Full harmonized urban-footprint series; 2005–2021 retained as a scale check
Fiscal appraisal / cadastre	Urban market-year	2010–2021	1,248	104 markets	104 Assessed-stock sums and means
WSF new built area	Urban market-year	1990–2015	2,704	104 markets	104 Physical built-form flow
VIIRS lights	Urban manzana-year	2013–2021	314,298	34,922 manzanas	104 Asinh mean radiance; mixed April-2012–March-2013 composite excluded
Manzana cadastre	Urban manzana-year	2010–2021	393,800	34,050 manzanas	104 Fiscal appraisal and cadastral stock; commerce share begins in 2012
Manzana WSF	Urban manzana-year	2010–2015	209,532	34,922 manzanas	104 Physical built-form flow
TDM proximity	Urban manzana	Static	34,922	34,922 manzanas	104 34.6 percent of manzanas within 10 km; 44.3 percent beyond 50 km
CONADI proximity	Urban manzana	Static	34,922	34,922 manzanas	104 Almost all manzanas within 40 km; 20–40 km used as reference
Mapuche share	Urban manzana	1992 / 2002 / 2017 snapshots	34,922	34,922 manzanas	104 Historical MDP/Redatam controls and 2017 zone-locality Mapuche share

Notes: The table reports the support used in the paper’s tables and figures. The counts differ across families because SII business data, cadastre, VIIRS lights, and WSF begin in different years and have different geographic support. Institutional-geography layers are attached to the fixed INE 2017 urban manzana geography; their temporal status is stated separately when they enter the analysis.

C Construction of the Analysis Data

The aggregate and manzana panels share identifiers and event classifications, but their outcomes do not have the same spatial support. This section records the construction choices that remain fixed across the main tables and appendix checks.

C.1 Urban systems, manzanas, and event assignment

The canonical geography begins with the 34,922 INE 2017 manzanas coded as urban (**AREA=1**) in the 104 study communes. Their union within each commune defines the fixed urban market. A 25-meter outward and inward closing in EPSG:32718 joins narrow streets and gaps without joining components that remain spatially separate. Interior holes are removed before recovering each component’s exterior edge. Event classification is global: an event inside any corrected component cannot enter the external exposure of another market.

Aggregate distance is measured from the event point to the corrected market polygon. Manzana distance is the minimum point-to-polygon distance; distance to the block centroid is retained only as a sensitivity. Inside events are first assigned by exact containment. Events in a street or corrected gap may be assigned to the nearest block only when they lie within 50 meters and in the same corrected component. The data retain exact, imputed, and unmatched counts. Direct events are separated from same-market spillovers at 0–1 and 1–5 kilometers and from events inside another urban market.

C.2 Administrative outcomes and linkage

SII enterprise counts and sales are commune-year tax-address outcomes. They are attached to the corresponding fixed urban market but cannot be restricted to its polygon. Fiscal-appraisal and physical-cadastre outcomes instead use the updated `catastral_new` parcel-coordinate bridge to the 2017 INE manzanas. The annual build combines fixed-width records for 2010–2011, S2 ZIP snapshots for 2012–2017, and delimited S2 records from 2018 onward. Linkage status and source era remain explicit because coverage and available fields change around 2017–2018.

C.3 Nighttime lights and built form

Aggregate luminosity is the harmonized DMSP–VIIRS DN product extracted with fractional-pixel weights to fixed urban-market polygons. The preferred main window ends in 2021 and contains a source-family transition in 2014. The audit therefore records low-light support, saturation, and sensor family by year. Manzana luminosity uses EOG VIIRS composites at approximately 500-meter resolution. Median polygon support is 0.032 pixels, and 63 percent of manzana-years cover less than 0.05 pixel. Exact extraction distinguishes polygon coverage but does not create independent block-resolution light measurements. The preferred specification consequently uses mean radiance from 2013–2021 and reports coarser-zone and extraction-method checks.

WSF Evolution records each pixel’s first built year. Annual new construction is derived from that field through 2015. The historical-support sensitivity uses only pixels built by 1989 to define a fixed core, baseline-unbuilt expansion zones, and alternative event-classification envelopes. This appendix exercise does not replace the INE 2017 geography.

C.4 Samples and transformations

Each outcome carries an explicit availability flag and estimation window. Transformations are fixed by measurement unit and zero support before the regressions are rerun: positive totals enter in logs, shares in percentage points, WSF flows as inverse-hyperbolic-sine hectares, and manzana VIIRS as inverse-hyperbolic-sine mean radiance. Alternative constants, levels, and count estimators are sensitivity checks rather than choices based on statistical significance.

C.5 Frozen outcome and treatment scales

Table A3 reports the prespecified scale checks generated from the same preferred controls and internal-conflict co-exposure as the main aggregate models. Panel A shows why the WSF result is not summarized by a single favorable transformation: the contemporaneous coefficient and the short-run dynamic response vary across levels, asinh, fixed-log-constant, and PPML specifications. Panel B changes the scale of the external-event treatment while holding the preferred outcome transformations fixed. These checks are sensitivities rather than a menu from which the most precise coefficient is selected.

Table A3: Outcome- and treatment-scale sensitivity

<i>Panel A. WSF outcome-scale sensitivity</i>				
Outcome scale	Static coef.	Static SE	Joint-lead p	Post 0–2 coef. (SE)
Preferred asinh hectares	0.012	0.032	0.376	-0.035 (0.014)
Hectares in levels	-1.053	0.438	0.441	-0.529 (0.577)
$\log(1+Y)$	0.008	0.026	0.457	-0.028 (0.012)
$\log(0.1+Y)$	0.007	0.043	0.224	-0.048 (0.017)
$\log(0.01+Y)$	-0.003	0.055	0.267	-0.059 (0.019)
PPML pixel count	-0.020	0.023	0.790	-0.021 (0.013)
<i>Panel B. Treatment-scale sensitivity: static coefficient (SE)</i>				
Treatment scale	SII sales	Enterprises	Urban NTL	WSF new
$\log(1+\text{event count})$	-0.019 (0.017)	-0.011 (0.006)	0.029 (0.011)	0.012 (0.032)
Event count in levels	0.002 (0.002)	-0.001 (0.001)	0.002 (0.002)	0.005 (0.005)
asinh event count	-0.016 (0.014)	-0.009 (0.005)	0.024 (0.009)	0.008 (0.025)
Any event	-0.031 (0.020)	-0.010 (0.007)	0.055 (0.016)	-0.044 (0.039)

Notes: Panel A holds the treatment at $\log(1 + \text{external event count})$. The preferred WSF outcome is asinh newly built hectares. The joint-lead column reports the covariance-aware Wald-test p-value; the final column reports the covariance-aware average over lags 0–2. Panel B holds each preferred outcome transformation fixed and varies the treatment scale. All specifications include market and year fixed effects, baseline-condition trends, a capital-status trend, and internal-conflict co-exposure. Standard errors are clustered by market.

D Formal-Activity Measurement and Sales Scale

Enterprise counts and tax-reported sales are registered at the commune tax address. They describe formal activity linked to an urban market, but they do not isolate transactions or firms operating inside the fixed urban footprint. The checks in this section make that limitation visible rather than treating the SII series as a direct measure of urban investment.

Table A4: Economic magnitude of the sales estimate

Contrast	Exposure change	Sales change (%)	Median-level equivalent
One-unit increase in exposure	1.000	-4.76	-114,108
Zero to one nearby event	0.693	-3.33	-79,684
Positive-exposure p25 to p75	1.540	-7.24	-173,498

Notes: The calculations use the final Panel A sales coefficient in Table 3, which conditions on annual own-market internal conflict with no distance cutoff. The median-level equivalent is expressed in the original SII sales units recovered from the logged outcome. The positive-exposure p25–p75 contrast is computed only among market-years with at least one broad-external event.

D.1 Sector composition of SII formal activity

Table A5 re-estimates the preferred aggregate specification separately for agriculture, forestry, all available SII *rubro* records, and the most relevant urban- and firm-facing sectors. Agriculture and forestry use the more detailed SII subrubro source; fishing and aquaculture are excluded. Sales and enterprise counts are lower in the all-rubro records. Agriculture sales decline, whereas forestry is imprecise; transport and storage declines on both margins. The exercise does not locate firms

inside the urban footprint. It checks composition rather than assigning the aggregate SII result to a single urban industry.

Figure A2 and Figure A3 report descriptive binned scatter plots between tax-reported sales and the three nighttime-lights measures. The raw plots show the expected cross-sectional relationship between formal sales and lights. The residualized plots remove market and year fixed effects and therefore better reflect the variation used in the panel specifications. The raw correlations are high, especially for urban-footprint lights, but the residualized correlations are weak: 0.10 for urban lights, 0.16 for whole-municipality lights, and 0.15 for the non-urban municipal residual. The figures are therefore a proxy diagnostic rather than a separate robustness table. Nighttime lights track the scale of markets in levels, but they do not closely measure the within-market formal activity margin identified by the panel estimates.

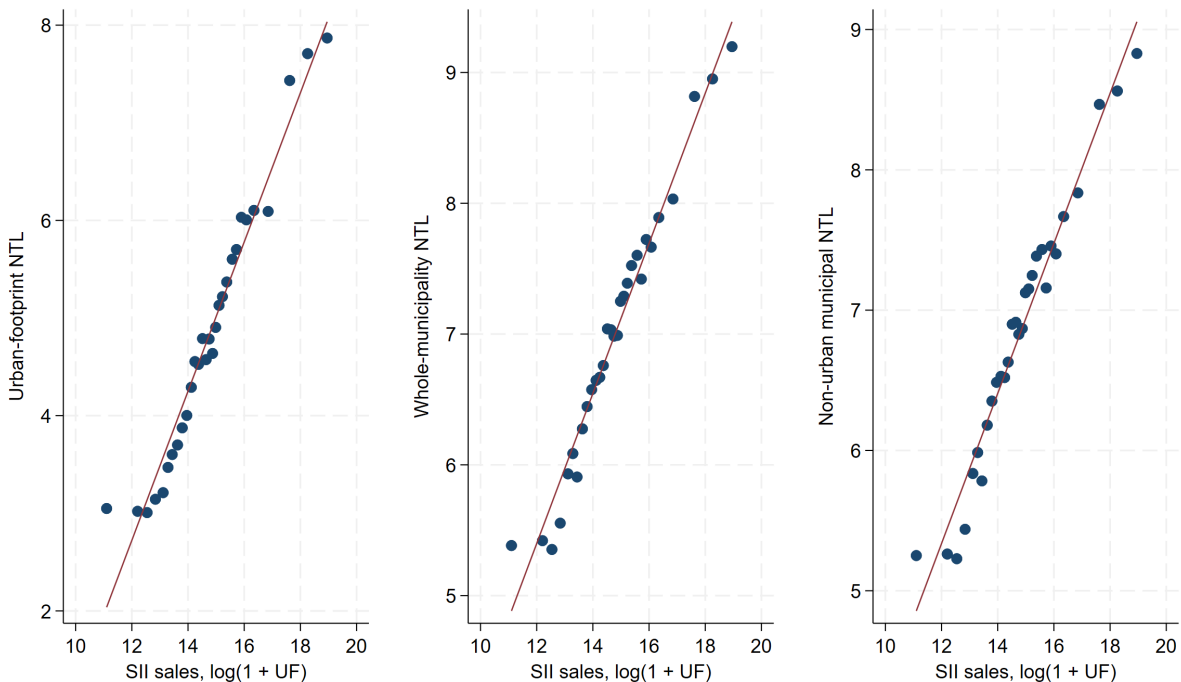


Figure A2: Relationship between tax-reported sales and nighttime lights. The figure plots binned means of log sales against urban-footprint, whole-municipality, and non-urban municipal harmonized nighttime lights in the common 2005–2021 sample.

Table A5: SII formal activity by rubro coverage and selected sector

Sector	Sales	Enterprises
Official SII total	-0.049*** (0.015)	-0.012** (0.005)
All SII rubro records	-0.032** (0.015)	-0.011** (0.005)
Agriculture	-0.162** (0.069)	-0.001 (0.008)
Forestry	0.045 (0.147)	0.001 (0.015)
Manufacturing	0.070 (0.091)	-0.005 (0.009)
Construction	0.276* (0.140)	0.022 (0.013)
Commerce	-0.020 (0.019)	-0.006 (0.005)
Transport and storage	-0.172** (0.081)	-0.021* (0.012)
Accommodation and food	0.216** (0.096)	0.004 (0.009)
Finance and insurance	-0.063 (0.147)	-0.051** (0.026)
Real estate	-0.247 (0.168)	-0.020 (0.021)

Notes: Coefficients use the final aggregate specification: broad external conflict within 30 kilometers; annual own-market conflict with no distance cutoff; market and year fixed effects; baseline-condition trends; and a capital-status trend. Standard errors are clustered by urban market. The all-rubro row pools every available SII rubro record. Agriculture and forestry use SII subrubro codes 011–017 and 021–024, respectively; fishing (031) and aquaculture (032) are excluded. The remaining sectors are selected and not exhaustive. SII outcomes remain commune tax-address measures. Protests are excluded.

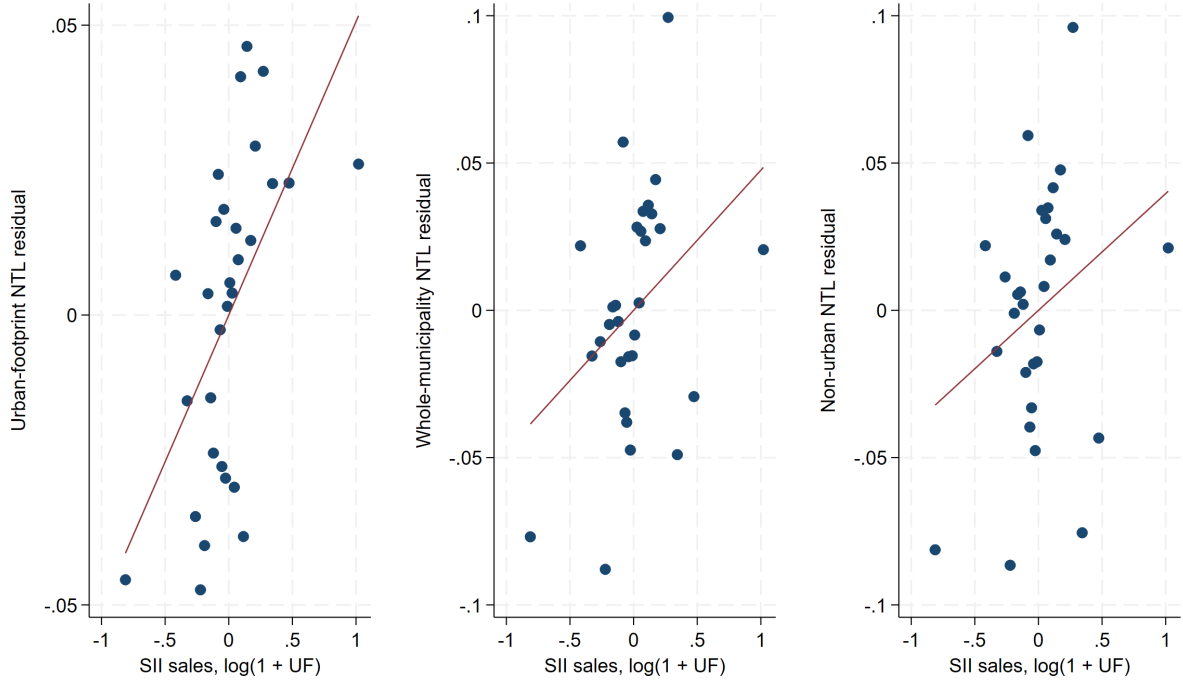


Figure A3: Residualized relationship between tax-reported sales and nighttime lights. Both sales and nighttime lights are residualized with respect to market and year fixed effects before binning.

E Local Exposure Geometry

Figure A8 shows nearby exposure around Temuco and Cañete. The wider view uses the 30-kilometer summary window. Red points mark broad external exposure: events outside every corrected urban system and events inside another urban market. Orange crosses mark events inside the receiving market, which enter the separate annual own-market co-exposure rather than the treatment.

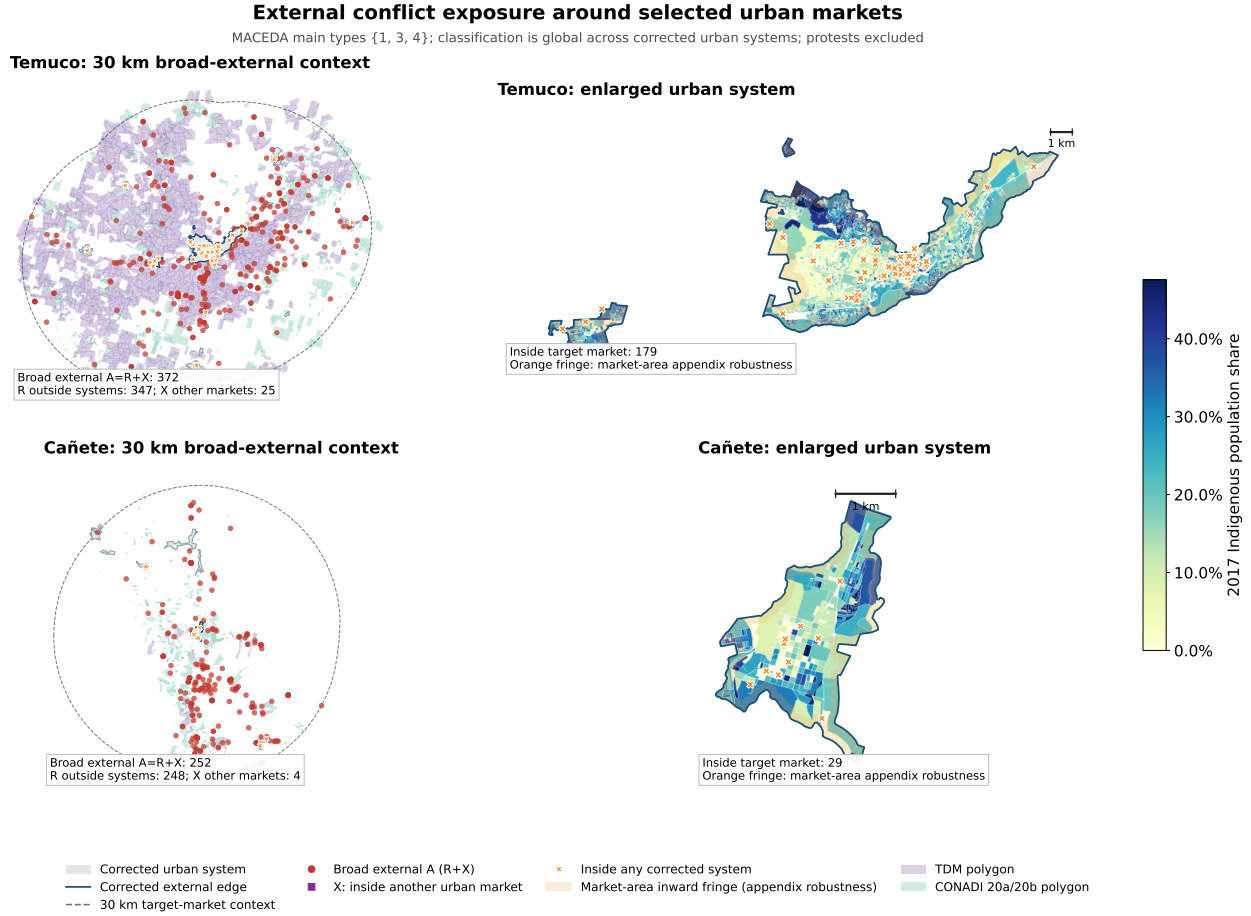


Figure A8: Local geometry of broad external conflict exposure in selected urban markets. Blue outlines show corrected urban-system edges. Red circles mark broad external events within 30 kilometers: events outside every corrected system (R) and events inside another urban market (X); purple squares identify the latter. Orange crosses mark main events inside an urban system. Enlarged panels retain the events assigned to the focal market, which form the separate annual own-market co-exposure. Shading denotes the 2017 Indigenous share, while TDM and CONADI polygons mark land-rights areas. Events are MACEDA types 1, 3, and 4 from 1990–2021; protests are excluded.

F Timing Diagnostics

F.1 Continuous timing checks

Figure 3 and Table 5 report the complete six-outcome continuous lead-lag evidence, including the three future-exposure terms and the covariance-aware averages over years 0–2, 3–5, and 6–8. Because nearby conflict is recurrent, these future terms diagnose prior movement but do not constitute a parallel-trends test.

F.2 Spike-style support audit

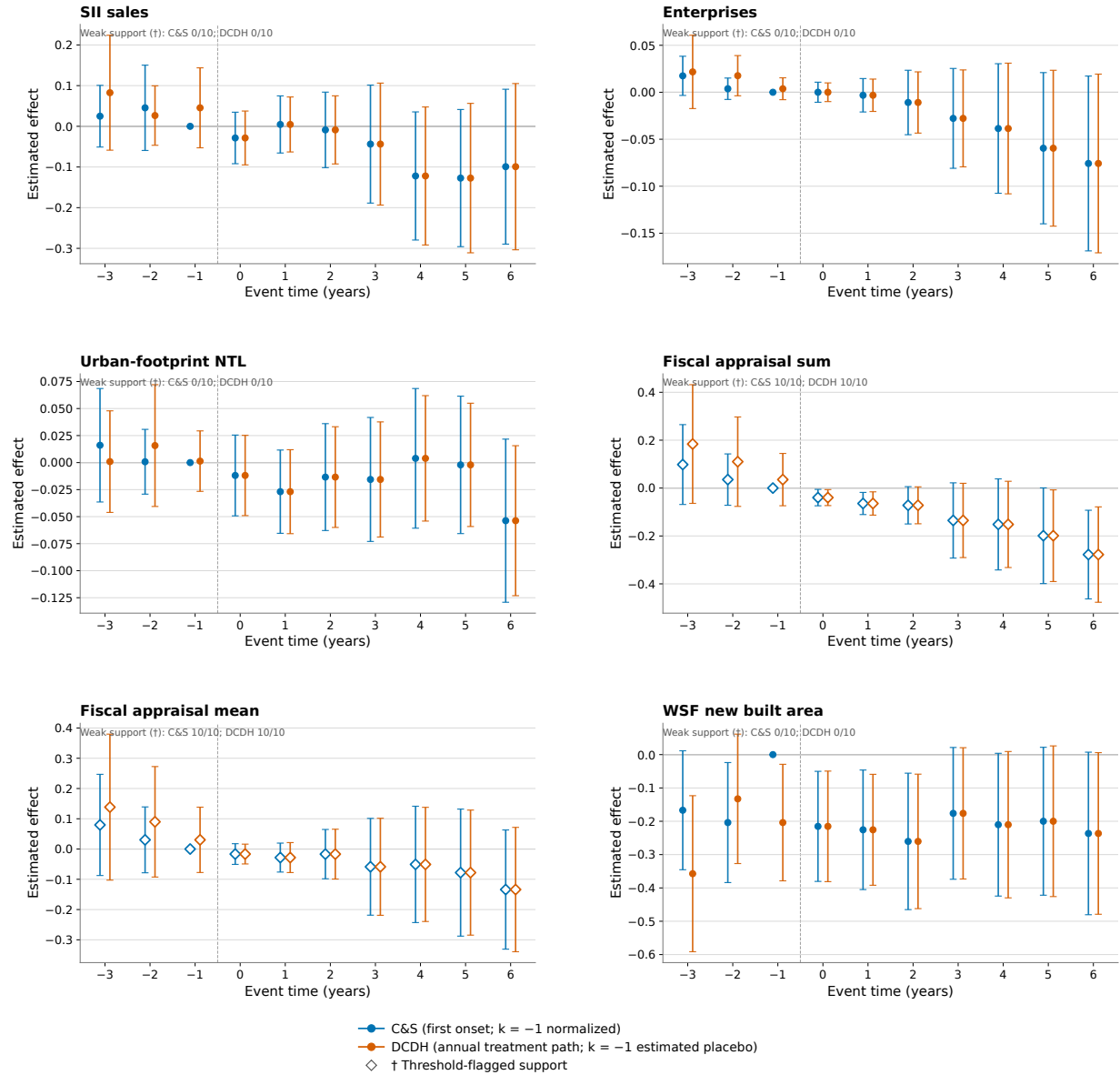
Figures A16–A18 hold exposure fixed at broad external conflict within 30 kilometers and use the same six outcomes. They display every estimable coefficient from event time $k = -3, \dots, +6$ as a point with a pointwise 95 percent confidence interval; the support flags remain visible rather than excluding flagged estimates.

The first coding uses a local-peak and extremeness rule inspired by the media-based unrest-event coding in Barrett et al. (2022). It is a project-specific annual spatial coding, not a replication of that media index or an absorbing treatment date. The remaining profiles use a cumulative p33 threshold adaptation and a recurrent annual p33 threshold sensitivity. The cumulative transformation is a translation of cumulative exposure considered by Neri-Lainé (2026), while the annual-increment transformation was created for this project; neither reproduces the source design or estimator.

For each coding, C&S group-time effects recode the first qualifying year as an absorbing post-first-spike assignment and compare cohorts with not-yet-treated markets (Callaway and Sant’Anna, 2021). The complementary de Chaisemartin–D’Haultfoeuille representation retains the annual binary path, so markets may return to zero or experience another spike. These are estimator choices made in this paper, not methods used by Barrett et al. or Neri-Lainé. They provide descriptive timing representations, not causal identification.

No outcome passes the joint support and placebo gate used for reporting these profiles. Some outcome families lack sufficient eligible first-spike markets; the better-supported nighttime-light and built-area samples fail the pre-period or placebo checks. Part of the local-peak coding also uses each market’s full 1990–2021 exposure distribution. The corridor peaks in 1999, 2009, 2016, and 2021 remain features of the descriptive conflict history in Figure 2; they do not define a separate treatment design.

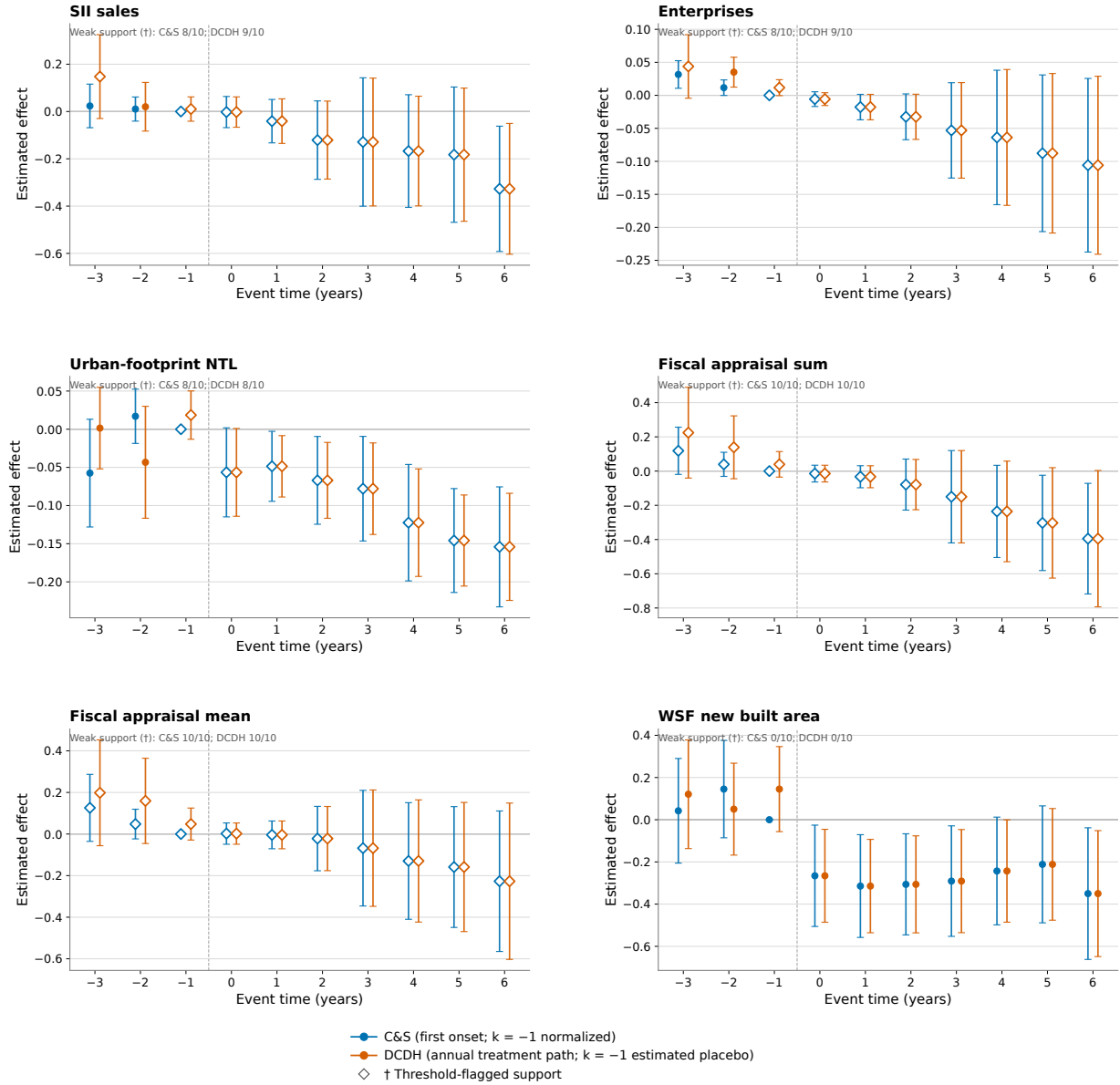
Barrett-inspired annual spike sensitivity: fixed 30 km broad-external conflict exposure



Notes: Each panel reports all estimable event-time coefficients for $k = -3, \dots, +6$ as points with pointwise 95% confidence intervals; coefficients are not connected. Exposure is broad external conflict ($A = R + X$) within 30 km; protests and own-market internal conflict are excluded. † flags terms below one or more diagnostic support thresholds (20 treated/switcher observations, 20 comparison observations, or 3 cohorts); flags do not remove estimates.

Figure A16: Fixed-window Barrett-inspired local-peak timing sensitivity. Each panel reports all estimable C&S first-onset and de Chaisemartin–D’Haultfoeulle annual-path coefficients for broad external conflict within 30 kilometers, over $k = -3, \dots, +6$. Points and bars are estimates and pointwise 95 percent confidence intervals; diamonds flag weak diagnostic support. The rule is a project-specific annual spatial coding inspired by Barrett et al. (2022), not a replication of its media-based index.

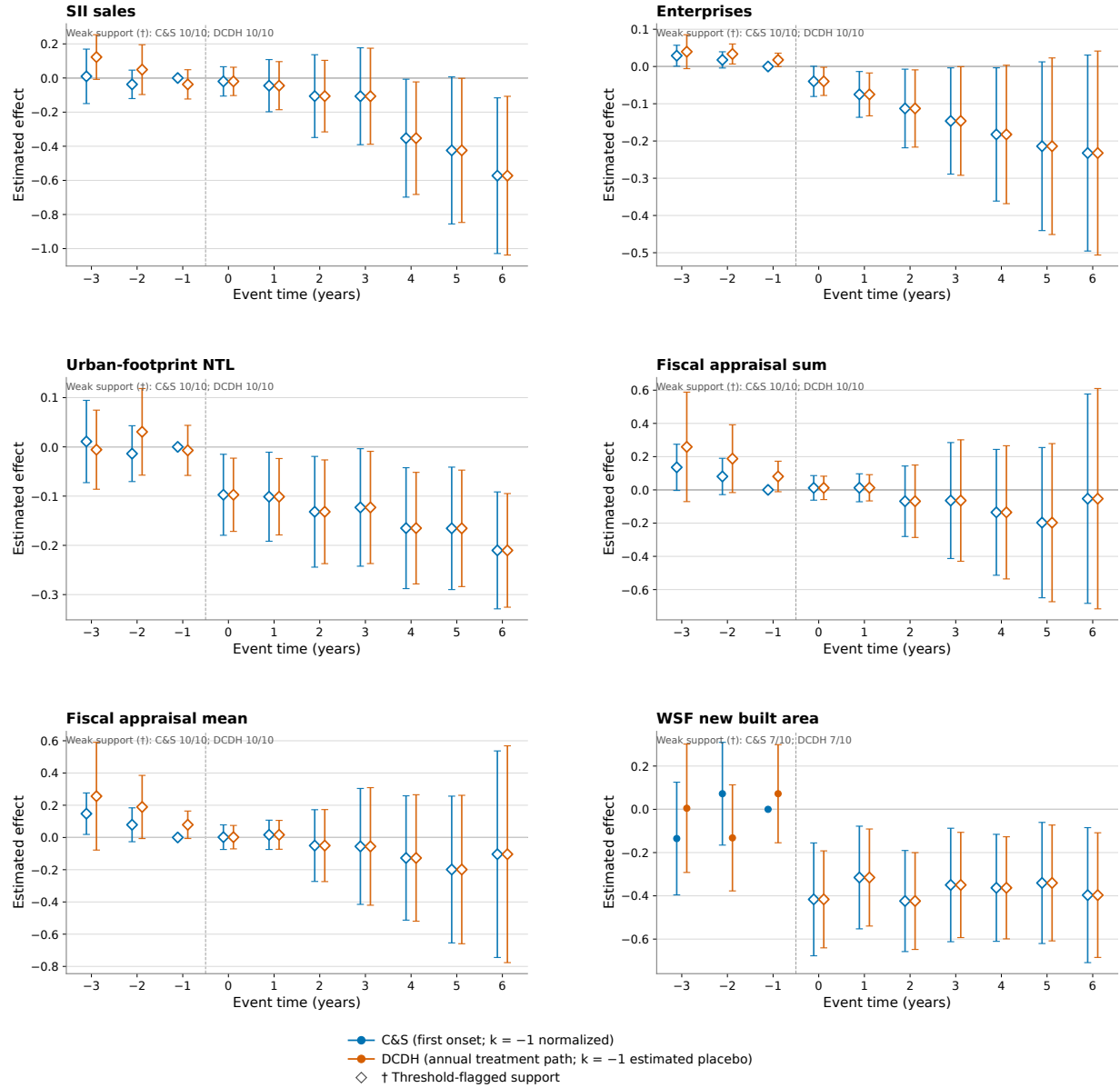
Cumulative p33 adaptation (inspired by Neri--Lainé): fixed 30 km broad-external conflict exposure



Notes: Each panel reports all estimable event-time coefficients for $k = -3, \dots, +6$ as points with pointwise 95% confidence intervals; coefficients are not connected. Exposure is broad external conflict ($A = R + X$) within 30 km; protests and own-market internal conflict are excluded. $\dagger$ flags terms below one or more diagnostic support thresholds (20 treated/switcher observations, 20 comparison observations, or 3 cohorts); flags do not remove estimates.

Figure A17: Fixed-window cumulative p33 threshold adaptation. Each panel reports all estimable C&S first-onset and de Chaisemartin–D’Haultfoeuille annual-path coefficients for broad external conflict within 30 kilometers, over $k = -3, \dots, +6$. Points and bars are estimates and pointwise 95 percent confidence intervals; diamonds flag weak diagnostic support. This is a project-specific threshold translation, not a replication of Neri–Lainé (2026).

Recurrent annual p33 sensitivity (project-created): fixed 30 km broad-external conflict exposure



Notes: Each panel reports all estimable event-time coefficients for $k = -3, \dots, +6$ as points with pointwise 95% confidence intervals; coefficients are not connected. Exposure is broad external conflict ($A = R + X$) within 30 km; protests and own-market internal conflict are excluded. † flags terms below one or more diagnostic support thresholds (20 treated/switcher observations, 20 comparison observations, or 3 cohorts); flags do not remove estimates.

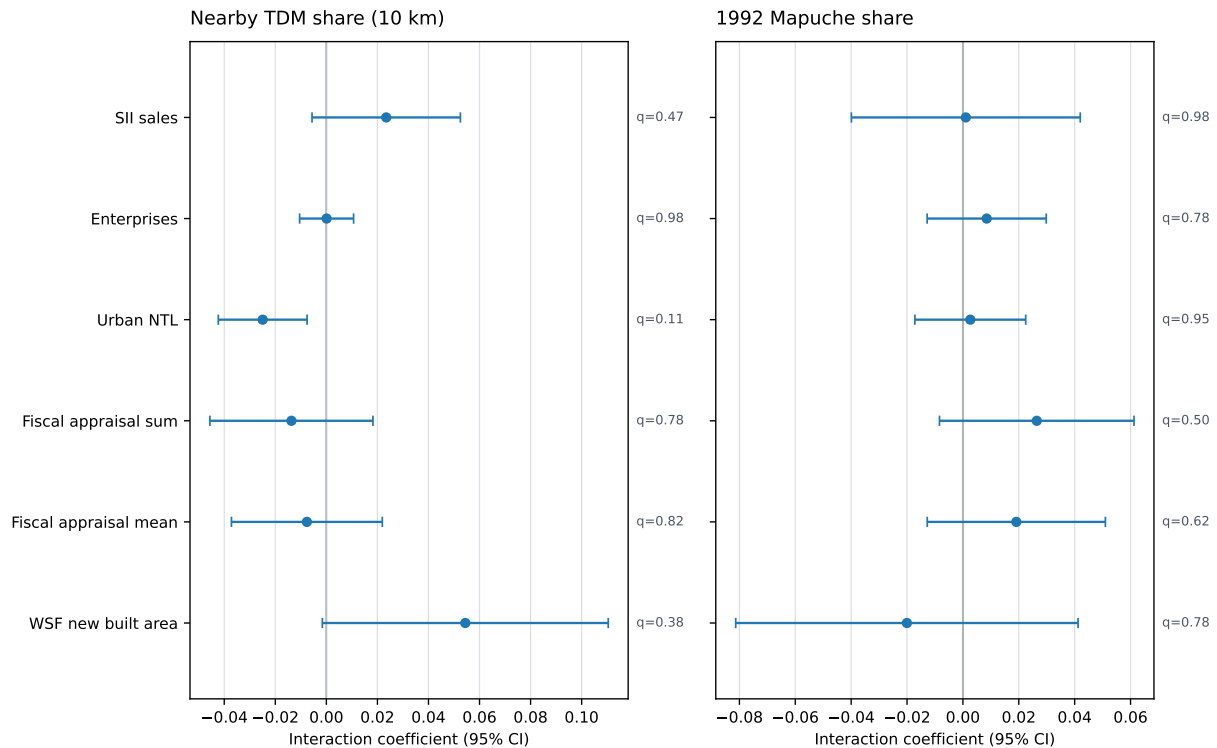
Figure A18: Fixed-window recurrent annual p33 threshold sensitivity. Each panel reports all estimable C&S first-onset and de Chaisemartin–D’Haultfoeuille annual-path coefficients for broad external conflict within 30 kilometers, over $k = -3, \dots, +6$. Points and bars are estimates and pointwise 95 percent confidence intervals; diamonds flag weak diagnostic support. The annual-increment rule was created for this project; it is not a Neri–Lainé rule or replication.

G Exploratory institutional geography

The main appendix heterogeneity exercise asks whether the broad-external association differs across historical claim geography and baseline Mapuche presence. It is deliberately narrower than a mechanism test. Nearby TDM share and the 1992 Mapuche population share are fixed before the panel outcomes, but they remain features of places that differ in many other respects.

Exploratory heterogeneity in broad-external conflict exposure

All models include the corresponding own-market-I interaction, preferred trends, market and year fixed effects; 95% CIs are market-clustered.



Both moderators are standardized across markets. BH q-values use the 12 pre-specified focal interactions and 999-replication wild-bootstrap p-values. None is below .05.

Figure A19: Exploratory heterogeneity by historical claim geography and 1992 Mapuche share. Each point is the interaction between broad external conflict within 30 kilometers and one standardized moderator; bars are market-clustered 95 percent confidence intervals. Both moderators enter jointly, together with their interactions with annual own-market conflict. The models otherwise use the preferred aggregate controls. The displayed q -values adjust the twelve pre-specified focal interactions using 999-replication wild-bootstrap p -values.

Table A6: Exploratory interactions with historical claim geography and 1992 Mapuche share

Outcome	Nearby TDM share (10 km)			1992 Mapuche share		
	β	Wild p	BH q	β	Wild p	BH q
SII sales	.023	.117	.468	.001	.958	.984
Enterprises	.000	.984	.984	.009	.458	.784
Urban NTL	-.025	.009	.108	.003	.796	.955
Fiscal appraisal sum	-.014	.409	.784	.026	.166	.498
Fiscal appraisal mean	-.008	.616	.821	.019	.260	.625
WSF new built area	.054	.064	.384	-.020	.523	.784

Notes: Each coefficient is the interaction between broad external conflict within 30 kilometers and a moderator standardized across markets. The TDM and Mapuche-share interactions enter jointly, as do their interactions with annual own-market conflict. All models include market and year fixed effects, baseline-condition trends, and a capital-status trend. Wild p -values use 999 market-clustered bootstrap replications. Benjamini–Hochberg q -values are calculated over all twelve focal interactions; none is below 0.05. These are exploratory differences across fixed geographies, not causal mechanism estimates.